IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Fan
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- **Methodologies:**
  - Collected data from SpaceX API and Wikipedia scraping;
  - wrangled for ML readiness;
  - EDA via visualizations and SQL;
  - interactive Folium maps and Plotly Dash;
  - tuned classification models (Logistic Regression, SVM, Decision Tree, KNN).
- **Key Results:**
  - Success rates climbed to 100% by 2020;
  - optimal payloads 2000-4000 kg;
  - SSO orbits at 100% success;
  - KSC LC-39A leads at 76.9%;
  - models achieved 83% test accuracy, with Decision Tree/SVM excelling in cross-validation.
- **Implications:** Reusability cuts costs to $62M vs. $165M competitors. SpaceY can target low-risk profiles (e.g., <4000 kg, coastal sites) for competitive bids.

Introduction

- **Background:** SpaceX's Falcon 9 reuses first stages, reducing launch costs significantly. Success depends on landings back to pads or drone ships.

- **Problem:** Predict landing success using factors like payload, orbit, site to estimate reuse and bid prices for SpaceY.

- **Data:** 90 launches (2010-2021) from API (structured details) and scraping (outcomes); features include mass, orbit (11 types), sites (4 coastal U.S.).

Section 1

Methodology

Methodology

Executive Summary

- **Data Collection:** API for launches/cores; scraping Wikipedia for versions/outcomes; merged on flight number.
- **Wrangling:** Imputed payload mean (3066 kg); binary success (1/0); one-hot encoded sites/orbits.
- **EDA:** Seaborn plots for trends; SQL aggregates on SQLite DB.
- **Interactive:** Folium for geospatial; Dash for filtered charts.
- **Modeling:** 80/20 split; GridSearchCV tuning; metrics: accuracy, precision, recall, F1, ROC-AUC.

Data Collection

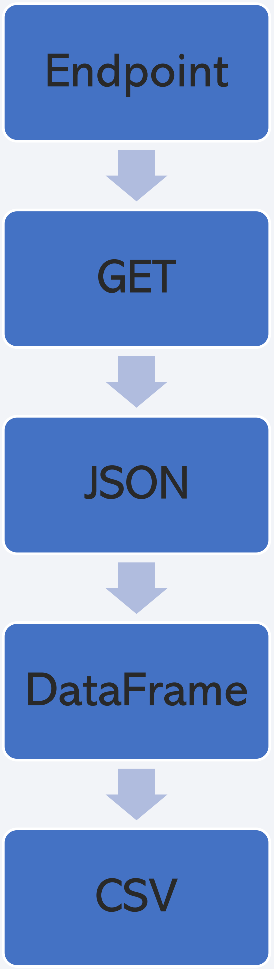
- **Process:**
  - Requests to API for JSON;
  - BeautifulSoup for Wiki tables.
  - Flowchart:

    API Query → Parse → Scrape → Merge → Clean.
- **Key:**
  - Extracted flight_number, site, mass, orbit, outcome;
  - handled 90 entries.

Data Collection – SpaceX API

- **Description:**
  - Fetched v4/launches;
  - fields: flight_number, launch_site, payload_mass_kg, orbit, core landing.

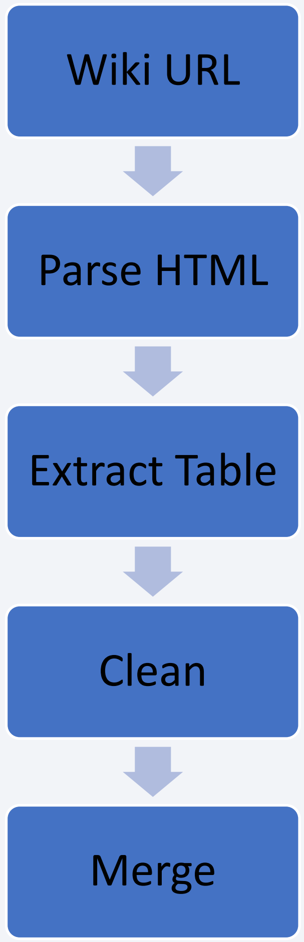
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/1.1_SpaceX_Data_Collection_API.ipynb



Data Collection - Scraping

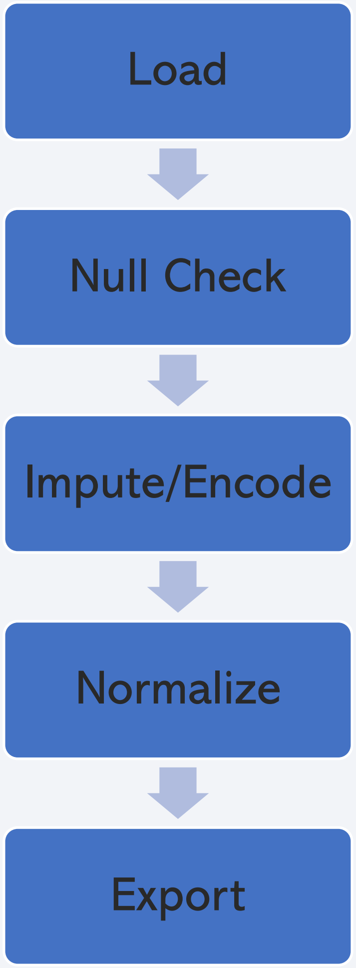
- **Description:**
  - Falcon 9 history table;
  - booster versions, binary outcomes.

- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/1.2_Web_Scraping.ipynb



Data Wrangling

- **Process:**
  - Dropped extras;
  - filled mass NaNs (mean 3066 kg);
  - encoded categoricals.
- **Key Phrases:**
  - Feature engineering for ML readiness;
  - handling imbalances in success/failure classes.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/1.3_SpaceX_Data_Wrangling.ipynb



EDA with Data Visualization

- **Charts Plotted:**
  - Scatter (Flight vs. Site) shows site shifts;
  - Scatter (Payload vs. Site) capacity limits;
  - Bar (Success vs. Orbit) SSO 100%;
  - Scatter (Flight vs. Orbit);
  - Scatter (Payload vs. Orbit);
  - Line (Yearly Success) from 0% in 2013 to 100% in 2020.
- **Rationale:**
  - Identified correlations low payloads/high success;
  - trends for features.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/2.2_EDA_Data_Visualization.ipynb

EDA with SQL

- **Queries Performed:**
  - Unique sites;
  - 'CCA' sites (5);
  - NASA payload (45,596 kg);
  - F9 v1.1 avg (2,928 kg);
  - First ground (2015-12-22);
  - Drone boosters 4000-6000 kg (F9 FT B1022, etc.);
  - Outcomes (98 success, 1 failure);
  - Max payload (F9 B5 B1048.4, etc.);
  - 2015 failures (F9 v1.1 B1012, etc.);
  - Ranked 2010-2017 (No attempt 10, etc.).

- **GitHub URL:** https://github.com/faansing/IBM-Data-Science/blob/main/2.1_EDA_SQL.ipynb

Build an Interactive Map with Folium

- **Map Objects:**
  - Markers (green success/red failure)
  - proximity lines to coast/rail/highway/city with distances (e.g., 0.86 km coast).
- **Rationale:**
  - Coastal site enable over-water paths;
  - proximities aid logistics/safety.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/3.1_Launch_Site_Location.ipynb

Build a Dashboard with Plotly Dash

- **Plots:**
  - Pie for site success;
  - scatter for payload vs. outcome with slider/filter.
- **Rationale:**
  - Dynamic views show KSC 76.9% success;
  - high rate 2000-4000 kg;
  - FT boosters strong.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/3.2_SpaceX_Dash_App.py

Predictive Analysis (Classification)

- **Process:**
  - Scaled features;
  - trained/tuned models;
  - all 83% test accuracy;
  - DT/SVM best in CV.
- **Flowchart:**
  - Split→ Train/Tune → Evaluate → Select.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/4.1_SpaceX_Machine_Learning_Prediction.ipynb

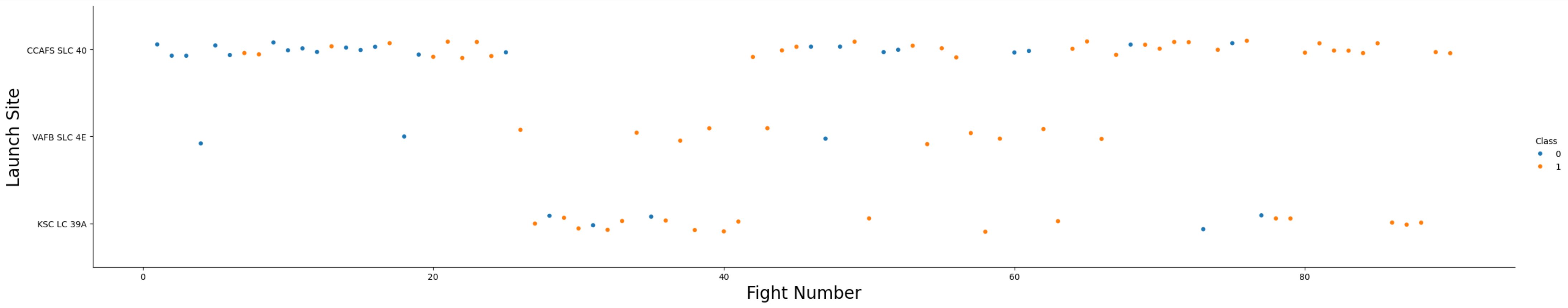
Results

- **EDA:**
  - Success improved yearly;
  - low payloads/orbits like SSO succeed more.
- **Interactive:**
  - Maps confirm coastal/logistics advantages;
  - dashboard pinpoints optimal ranges.
- **Predictive Results:**
  - 83% accuracy;
  - No false negatives in best models – key for risk.

Section 2

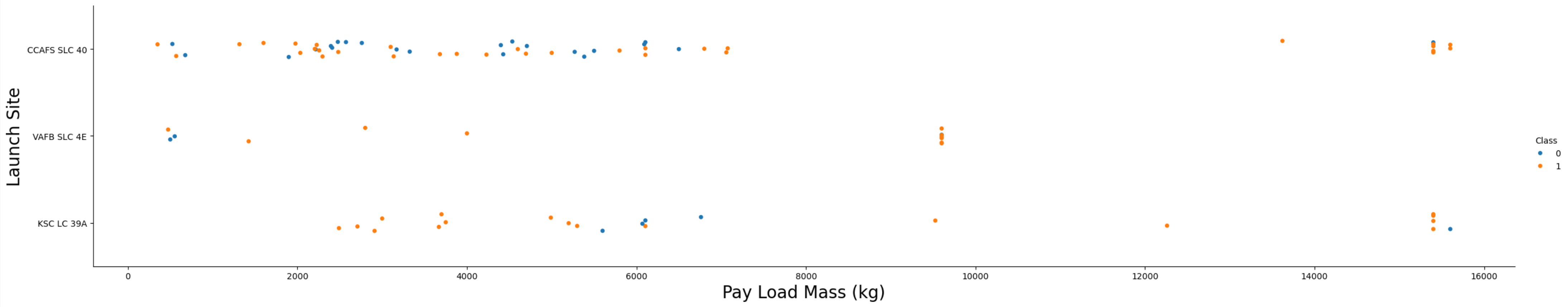
Insights drawn from EDA

Flight Number vs. Launch Site



- Explanation:
  - CCAFS early mix;
  - KSC consistent post-2017.

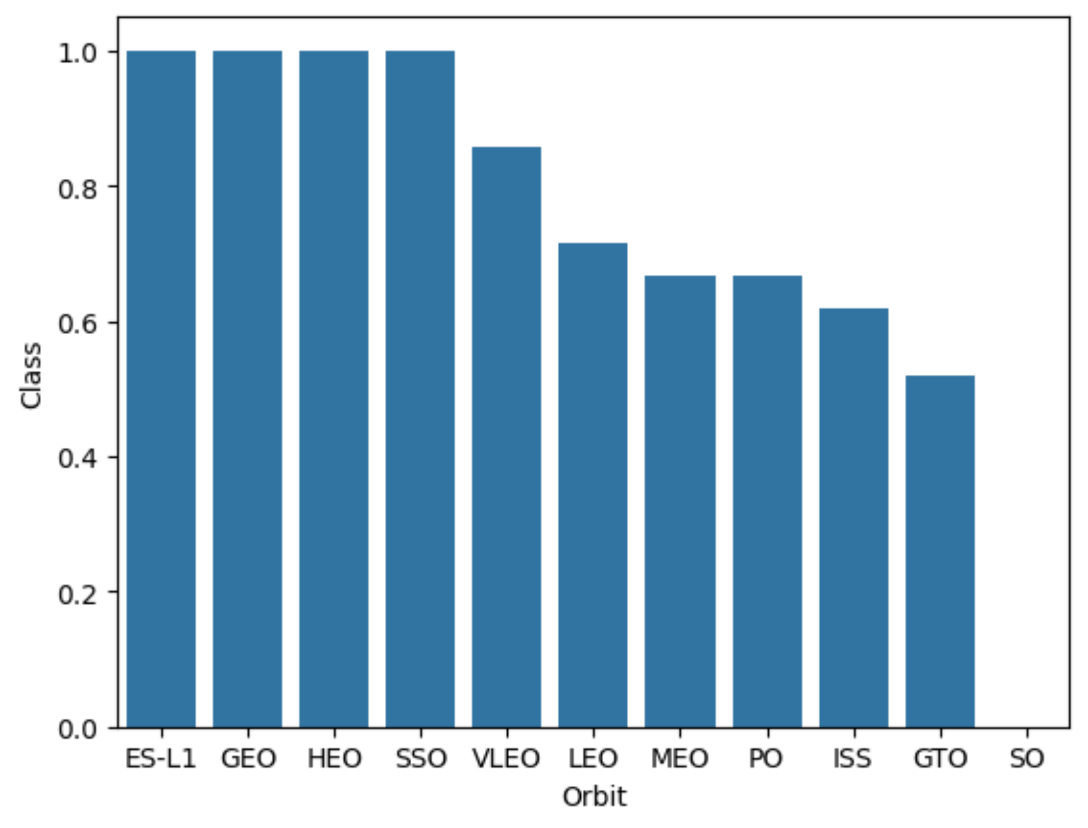
Payload vs. Launch Site



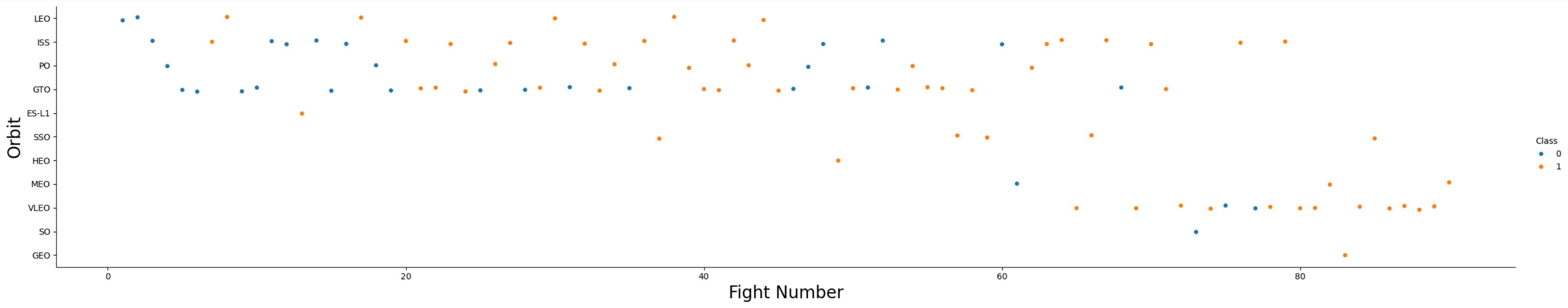
- Explanations:
  - VAFB light loads high success;
  - CCAFS heavy risks.

Success Rate vs. Orbit Type

- Explanation:
  - Energy demands lower GTO rates.

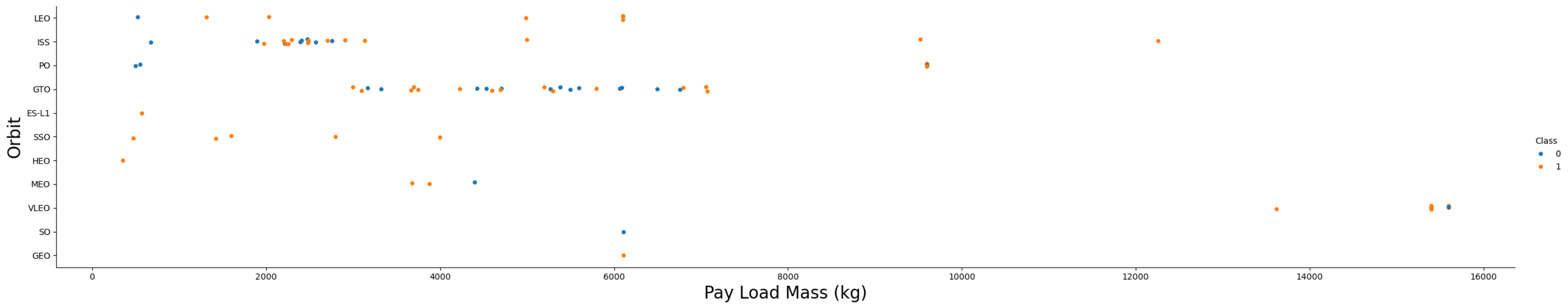


Flight Number vs. Orbit Type



- Explanation:
  - Recent VLEO/ISS improvements.

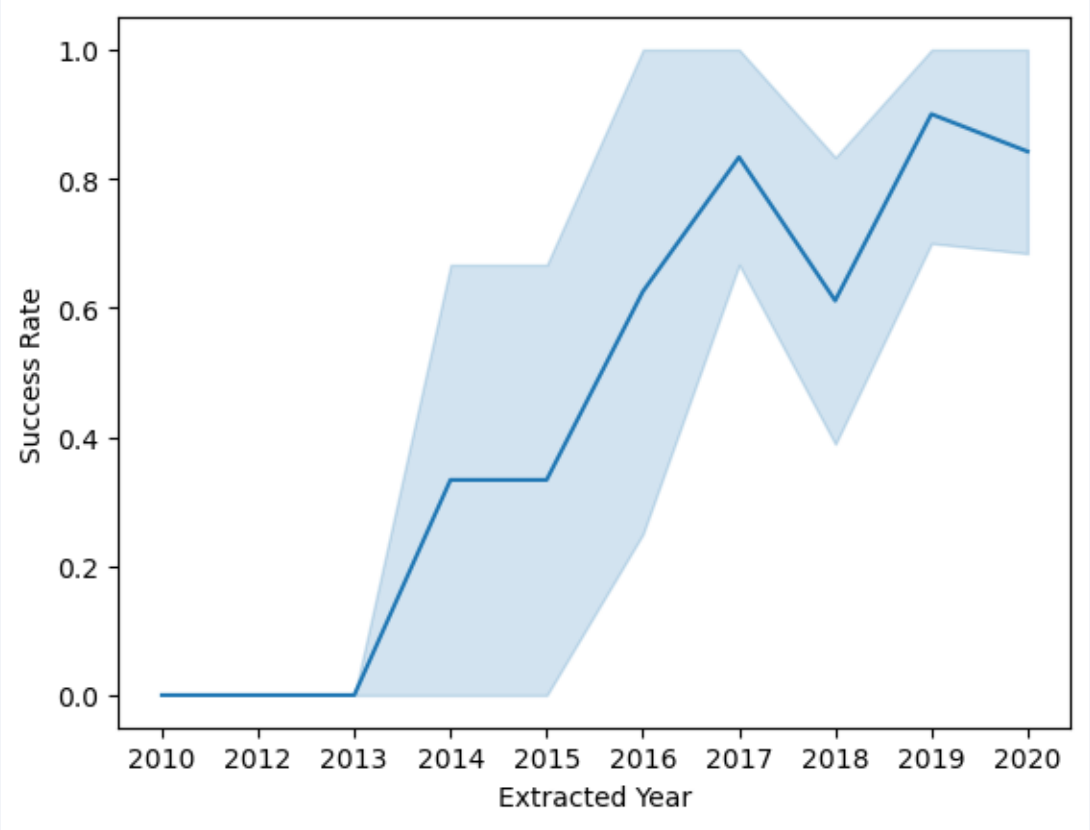
Payload vs. Orbit Type



- Explanation:
  - 2000-4000 kg optimal;
  - heavy GTO fails.

Launch Success Yearly Trend

- Explanation:
  - Tech maturation drives gains.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

In [11]:

```
## Downgrad `prettytable` to version 3.5.0 to ensure compatibility with ipython-sql-0.5.0

!pip install prettytable==3.5.0
```

```
Requirement already satisfied: prettytable==3.5.0 in /opt/conda/lib/python3.12/site-packages (3.5.0)
Requirement already satisfied: wcwidth in /opt/conda/lib/python3.12/site-packages (from prettytable==3.5.0) (0.2.1
3)
```

In [12]:

```
%sql SELECT DISTINCT launch_site FROM SPACEXTBL
```

```
 * sqlite:///my_data1.db
Done.
```

Out[12]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Explanation:
  - Four U.S. coastal bases.

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

In [13]:
```
%sql SELECT * FROM SPACEXTBL WHERE launch_site LIKE 'CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (p
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (p
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	N
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	N
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	N

- Explanation:
  - Cape Canaveral subset.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

In [14]:

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE customer LIKE 'NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

Total_Payload_Mass
45596

- Explanation:
  - NASA's SpaceX cargo total.

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
In [15]: %sql SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1'

         # 'F9 v1.1  B1003', 'F9 v1.1', 'F9 v1.1 B1011', 'F9 v1.1 B1010', 'F9 v1.1 B1012', 'F9 v1.1 B1013', 'F9 v1.1 B1014

 * sqlite:///my_data1.db
Done.
```

Out[15]:

Avg_Payload_Mass
2928.4

- Explanation:
    - Early version benchmark.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

In [16]:
```
%sql SELECT MIN(Date) AS Date FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Date
2015-12-22

- Explanation:
  - Reusability breakthrough

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

In [17]:
```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE (Landing_Outcome = 'Success (drone ship)') AND (PAYLOAD_MASS__KG
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Explanation:
  - Mid-payload ocean successes.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

In [18]:
```
%sql SELECT Mission_Outcome, COUNT(*) AS Count FROM SPACEXTBL GROUP BY Mission_Outcome ORDER BY Mission_Outcome


# %sql SELECT COUNT(Mission_Outcome) AS Sueecss_Outcome FROM SPACEXTBL WHERE Mission_Outcome LIKE 'Success%'
# %sql SELECT COUNT(Mission_Outcome) AS Failure_Outcome FROM SPACEXTBL WHERE Mission_Outcome LIKE 'Failure%'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Explanation:
  - High reliability overall.

Boosters Carried Maximum Payload

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

In [19]:
```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ IN (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXT
```

```
 * sqlite:///my_data1.db
Done.
```

Out[19]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

- Explanation:
  - Heavy-lift performers at 15,600 kg.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

In [20]:
```
%sql SELECT SUBSTR(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE SUBST
```

```
 * sqlite:///my_data1.db
Done.
```

Out[20]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Explanation:
  - Early drone challenges.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [21]:

```
%sql SELECT Landing_Outcome, COUNT(Landing_Outcome) AS Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '
# %sql SELECT LANDING__OUTCOME FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' ORDER BY DATE DESC
```

```
 * sqlite:///my_data1.db
Done.
```

Out[21]:

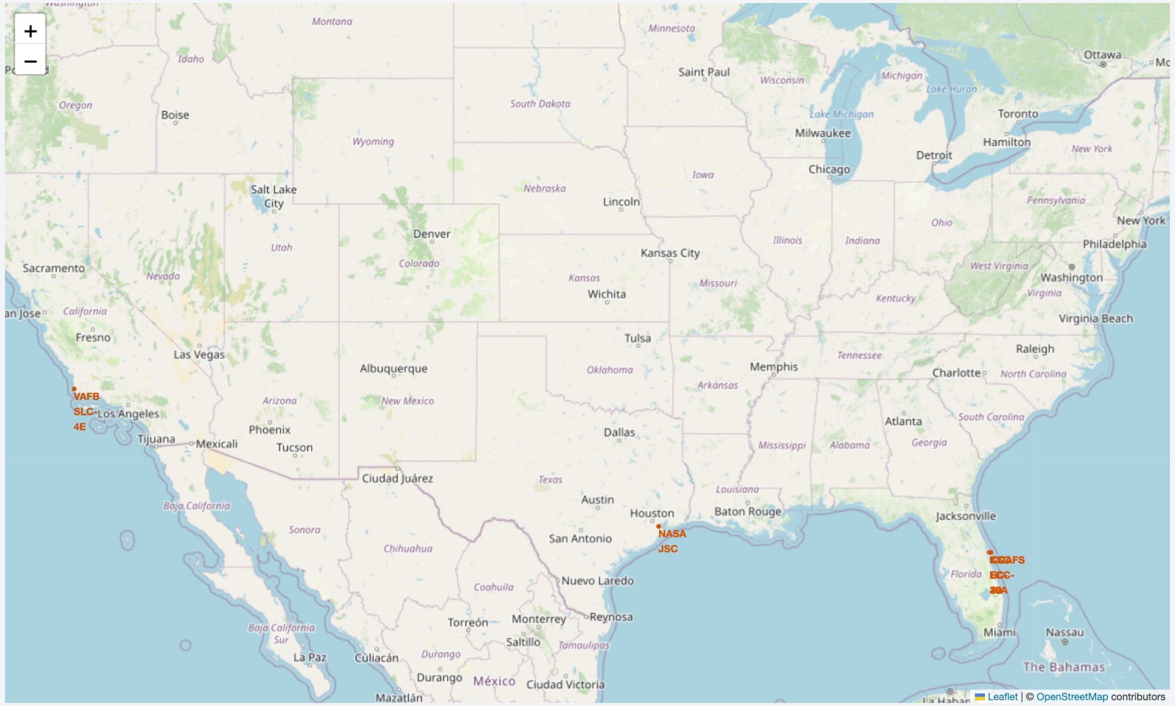
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Explanation:
  - Evolution from no attempts to balanced outcomes.

Section 3

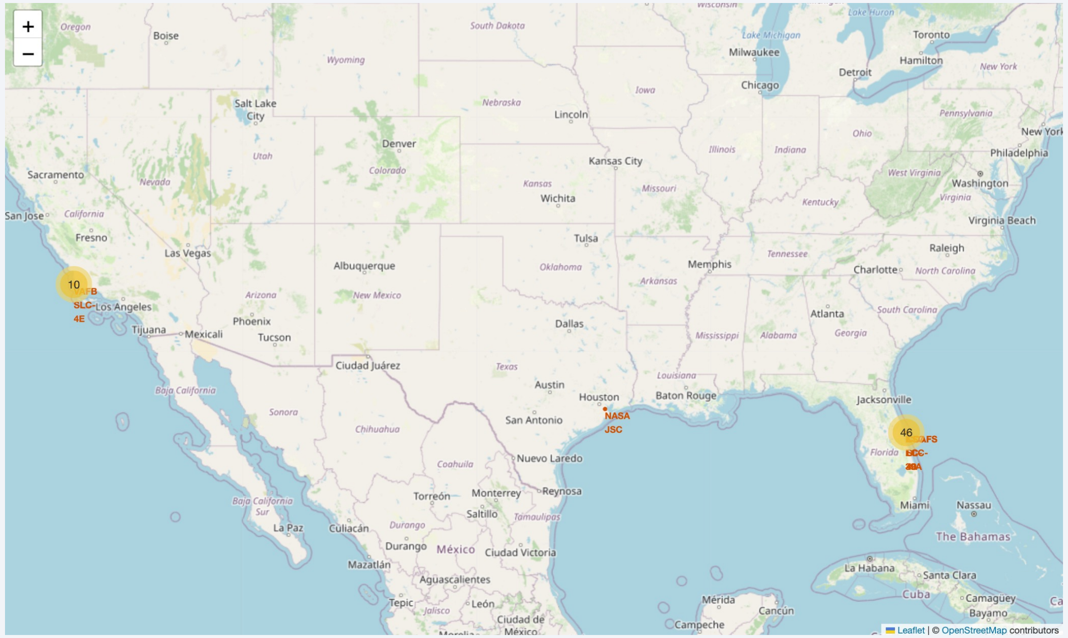
Launch Sites Proximities Analysis

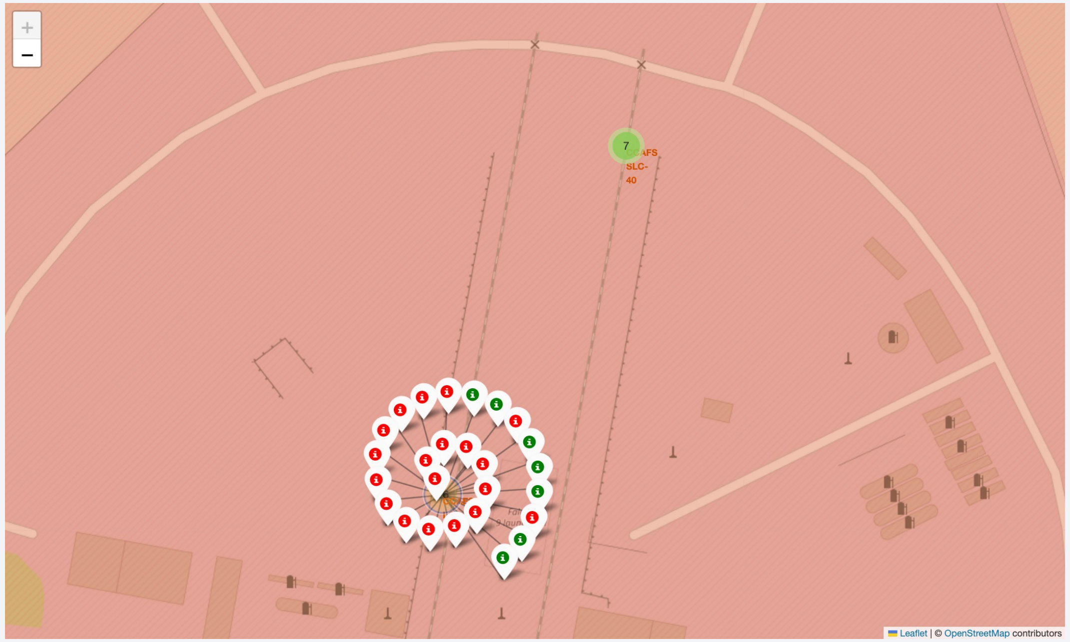
Folium Map: Global Launch Sites



- Explanation:
    - Clustered on coasts for safe trajectories.

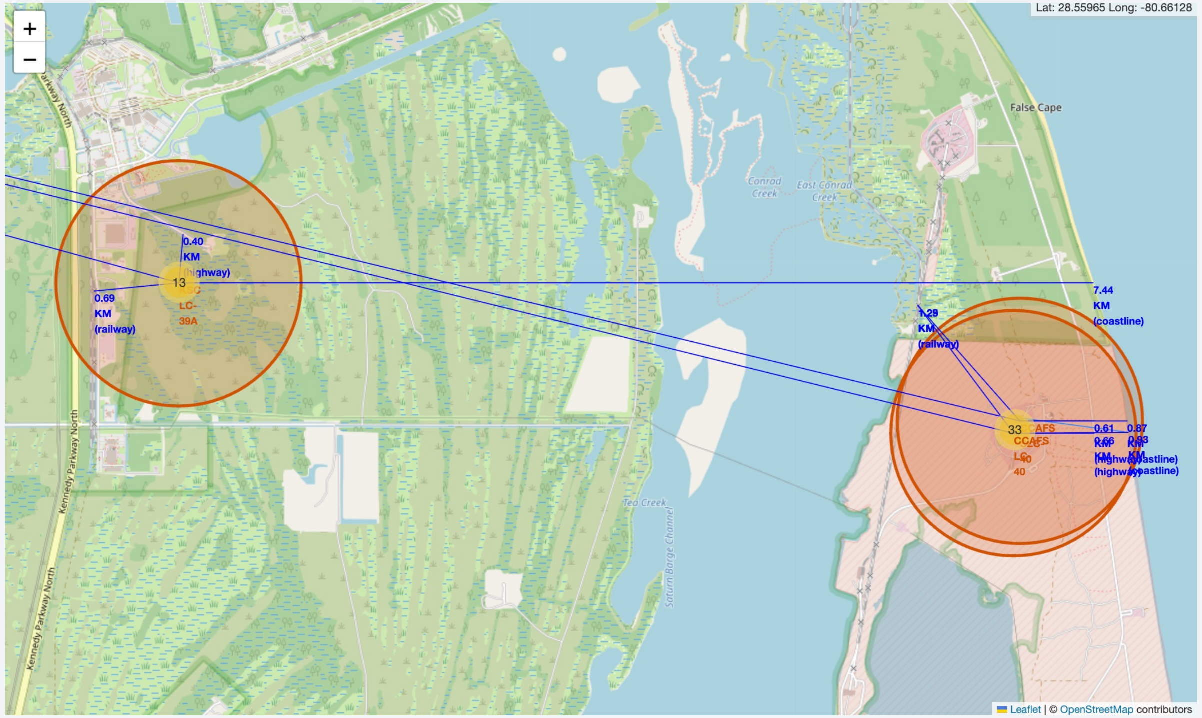
Folium Map: Launch Outcomes





- Explanation:
  - Successes near coasts;
  - failures offshore.

Folium Map: Site Proximities

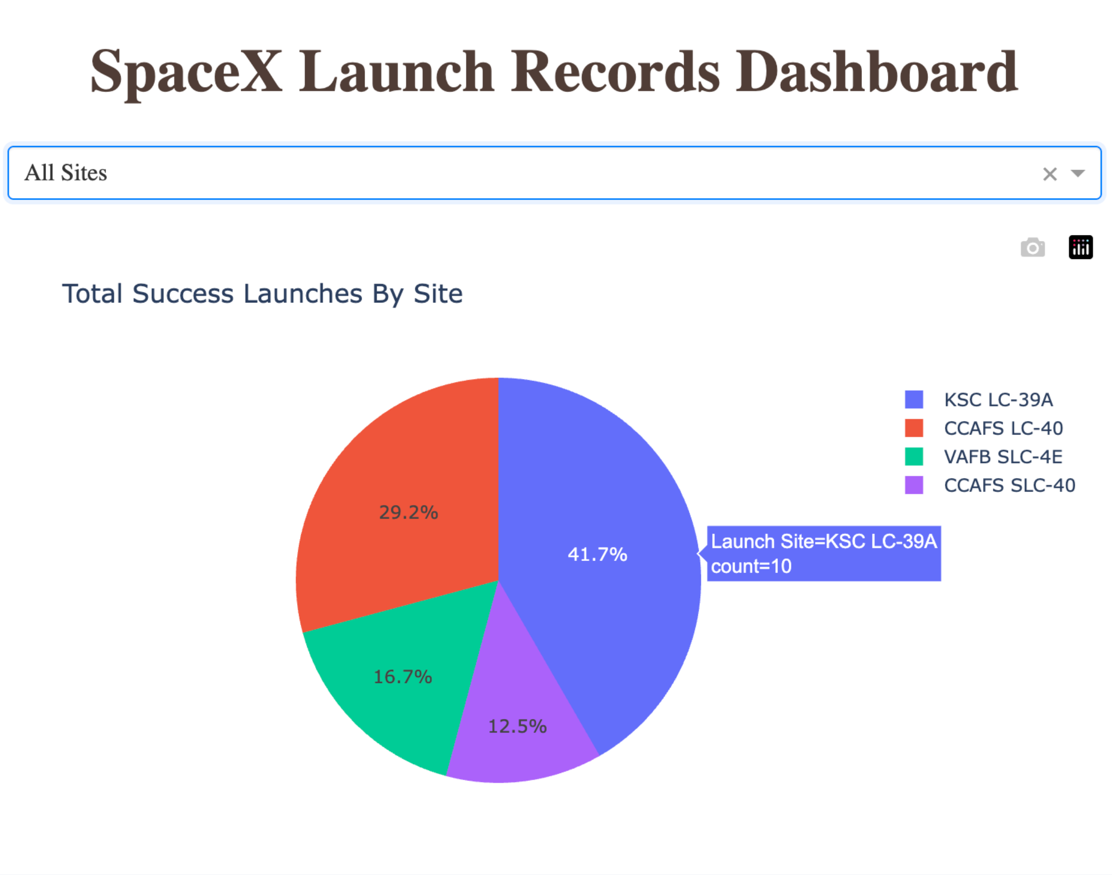


- Explanation:
  - Balances access and safety.

Section 4

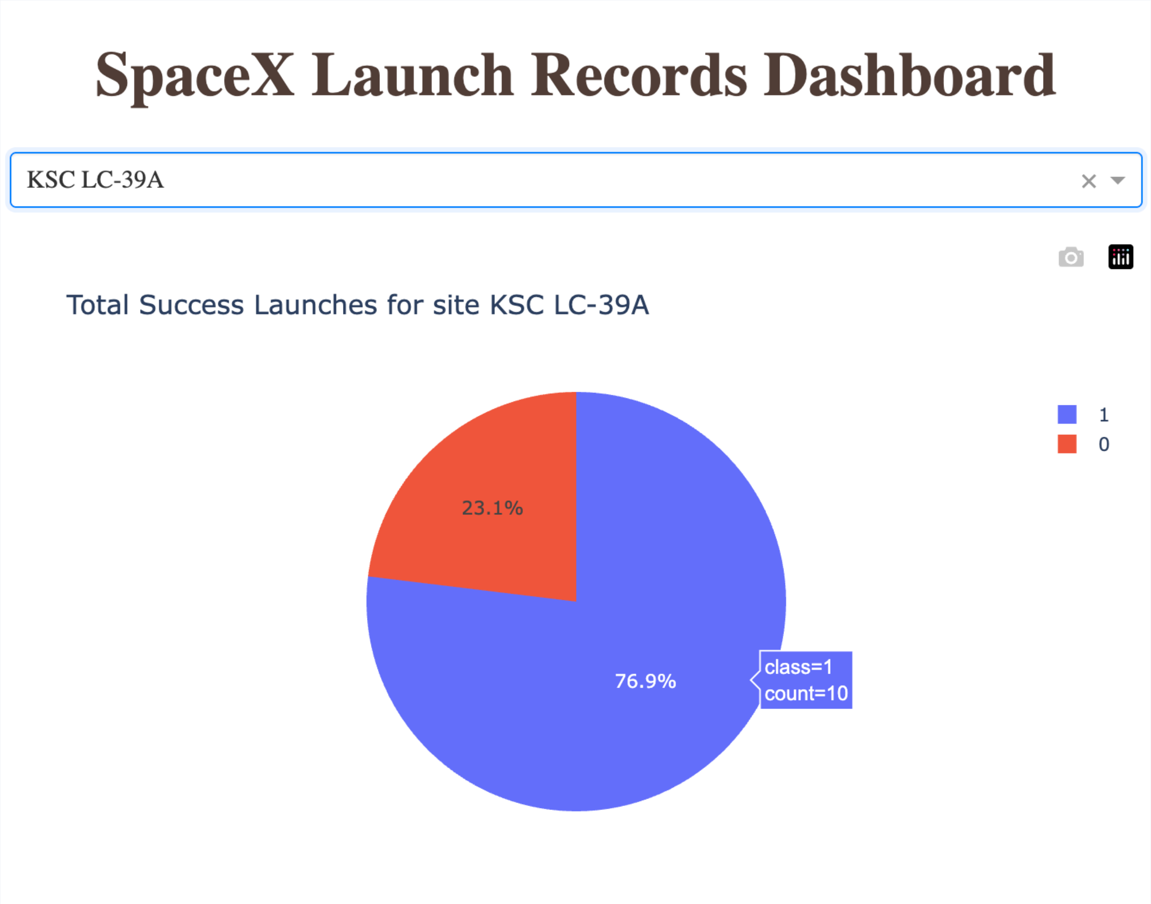
Build a Dashboard with Plotly Dash

Dashboard: Launch Success by Site



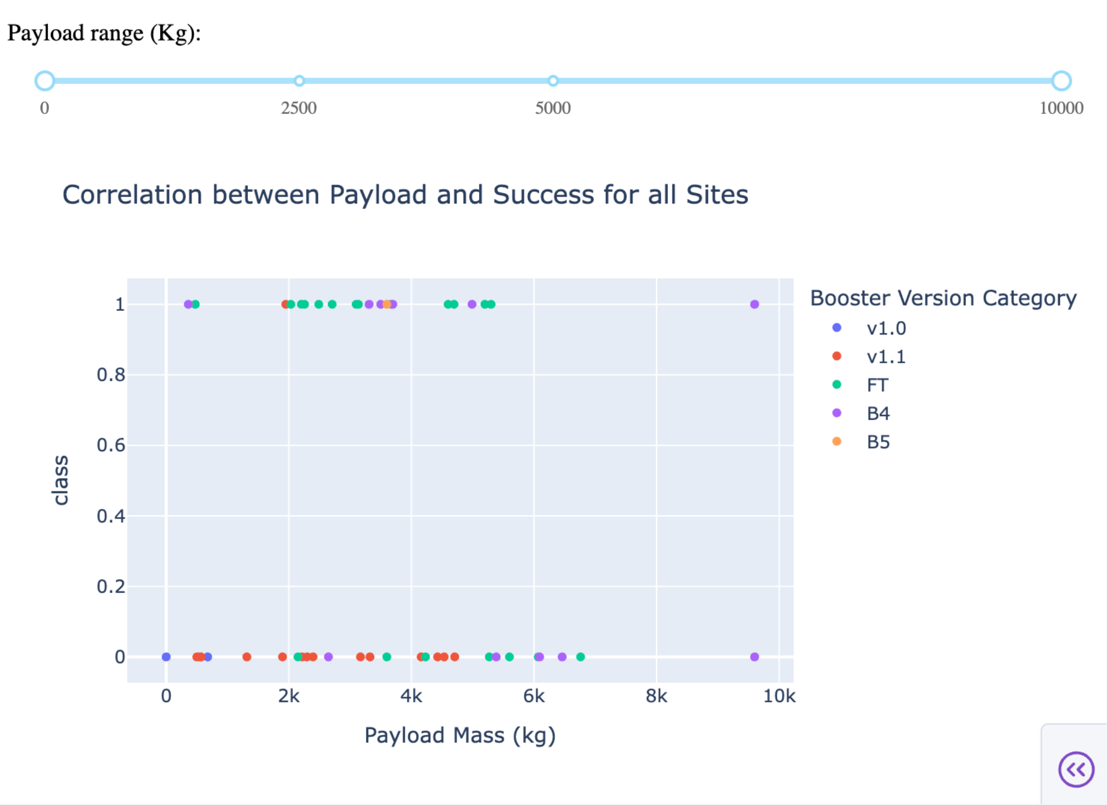
- Explanation:
  - KSC leads absolute successes.

Dashboard: Highest Success Site



- Explanation:
    - Modern infrastructure boosts rate.

Dashboard: Payload vs. Outcome



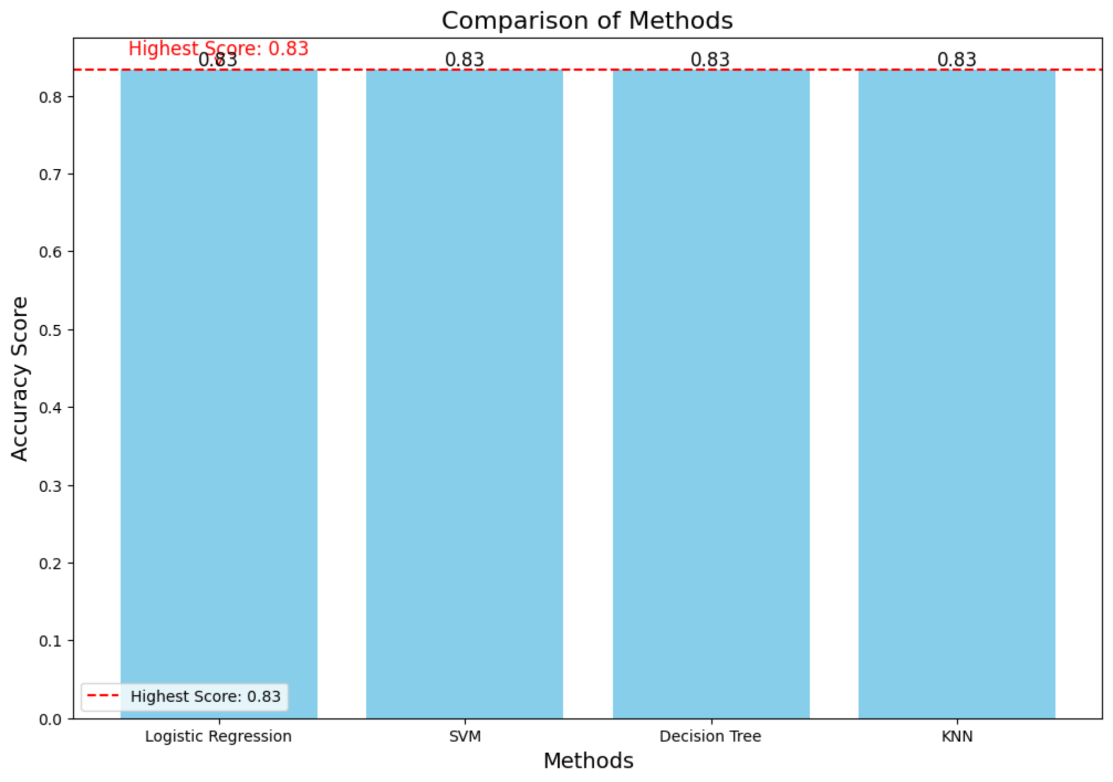
- Explanation:
  - Drops above 6000 kg;
  - later versions improve.

Section 5

Predictive Analysis (Classification)

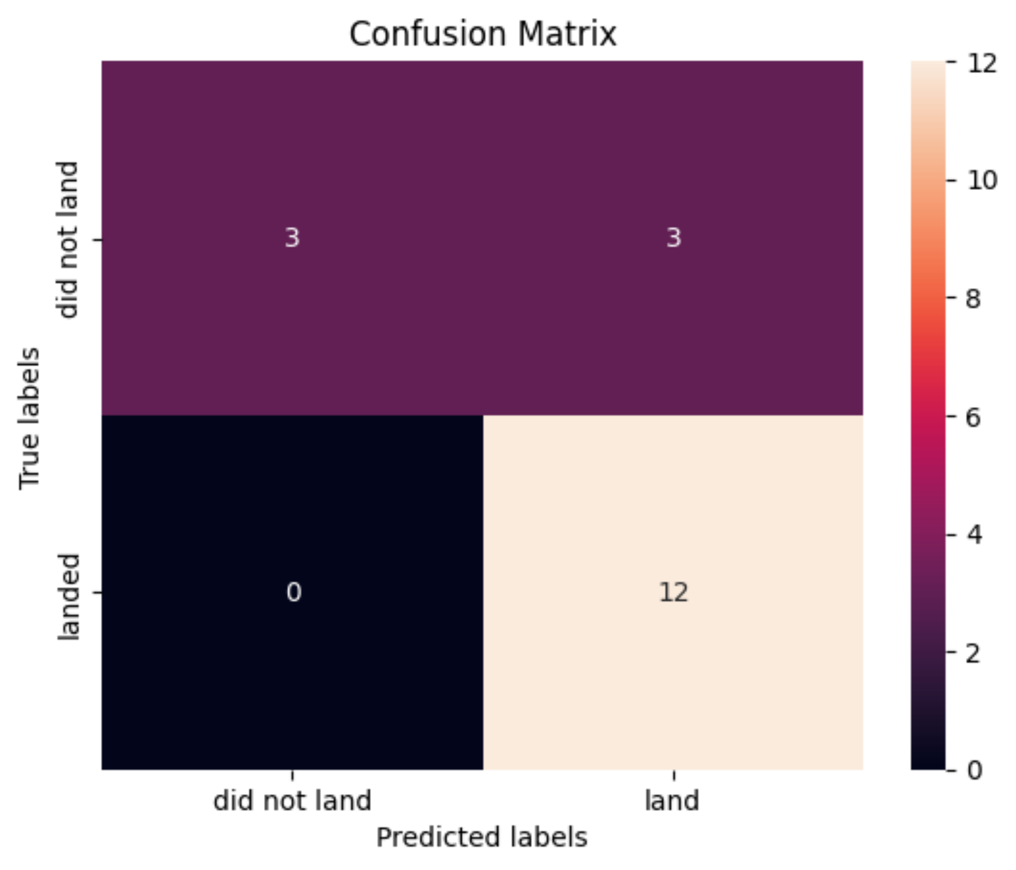
Classification Accuracy

- Bar Chart:
  - All models 83% on test;
  - DT/SVM higher in CV.
- Best Model:
  - Decision Tree (balanced metrics).



Confusion Matrix

- Matrix for Best:
  - TP 12, TN 3, FP 3, FN 0
- Explanation:
  - No missed failures;
  - conservative on risks.



Conclusions

- **Trends:**
  - Successes rose with experience;
  - favor low payloads, SSO/GTO orbits, KSC.
- **Predictors:**
  - Orbit, mass, site key for 83% accuracy.
- **For SpaceY:**
  - Prioritize <4000 kg missions;
  - mimic coastal setups.
- **Next:**
  - Integrate real-time weather;
  - test on new data.

Appendix

- Reference:
  - https://www.coursera.org/learn/applied-data-science-capstone/
  - https://github.com/faansing/IBM-Data-Science/

Thank you!